Machine-Level Programming I: Basics

15-213/18-213/15-213: Introduction to Computer Systems
5th Lecture, September 12, 2017

Today's Instructor:

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程序的转换与机器级表示

- 主要教学目标
  - 了解高级语言与汇编语言、汇编语言与机器语言之间的关系
  - 掌握有关指令格式、操作数类型、寻址方式、操作类型等内容
  - 了解高级语言源程序中的语句与机器级代码之间的对应关系
  - 了解复杂数据类型（数组、结构等）的机器级实现
- 主要教学内容
  - 介绍C语言程序与机器级指令之间的对应关系。
  - 主要包括：程序转换概述、IA-64指令系统、C语言中控制语句和过程调用等机器级实现、复杂数据类型（数组、结构等）的机器级实现等。
  - 本章所用的机器级表示主要以汇编语言形式表示为主。

采用逆向工程方法！

程序的机器级表示

- 分以下五个部分介绍
  - 第一讲：程序转换概述
    - 机器指令和汇编指令
    - 机器级程序员感觉到的属性和功能特性
    - 高级语言程序转换为机器代码的过程
  - 第二讲：IA-32 /x86-64指令系统
  - 第三讲： C语言程序的机器级表示
    - 选择语句的机器级表示
    - 循环结构的机器级表示
    - 过程调用的机器级表示
  - 第四讲：复杂数据类型的分配和访问
    - 数组的分配和访问
    - 结构体数据的分配和访问
    - 联合体数据的分配和访问
    - 数据的对齐
  - 第五讲：越界访问和缓冲区溢出

从高级语言程序出发，用其对应的机器级代码以及内存（栈）中信息的变化来说明底层实现

围绕C语言中的语句和复杂数据类型，解释其在底层机器级的实现方法

“指令”的概念

- 计算机中的指令有微指令、机器指令和伪（宏）指令之分
- 微指令是微程序级命令，属于硬件范畴
- 伪指令是由若干机器指令组成的指令序列，属于软件范畴
- 机器指令介于二者之间，处于硬件和软件的交界面
  - 本章中提及的指令都指机器指令
- 汇编指令是机器指令的汇编表示形式，即符号表示
- 机器指令和汇编指令一一对应，它们都与具体机器结构有关，都属于机器级指令

机器级指令

- 机器指令和汇编指令一一对应，都是机器级指令
- 机器指令是一个0/1序列，由若干字段组成

补码11111010的真值为多少？

100010 DW	mod	reg	r/m	disp8
100010 0 0	01	001	001	11111010

操作码　寻址方式　寄存器编号　立即数(位移量)

- 汇编指令是机器指令的符号表示（可能有不同的格式）

mov [bx+di-6], cl　或　movb %cl, -6(%bx,%di)

Intel格式　AT&T 格式

mov、movb、bx、%bx等都是助记符

指令的功能为：M[R[bx]+R[di]-6]←R[cl]

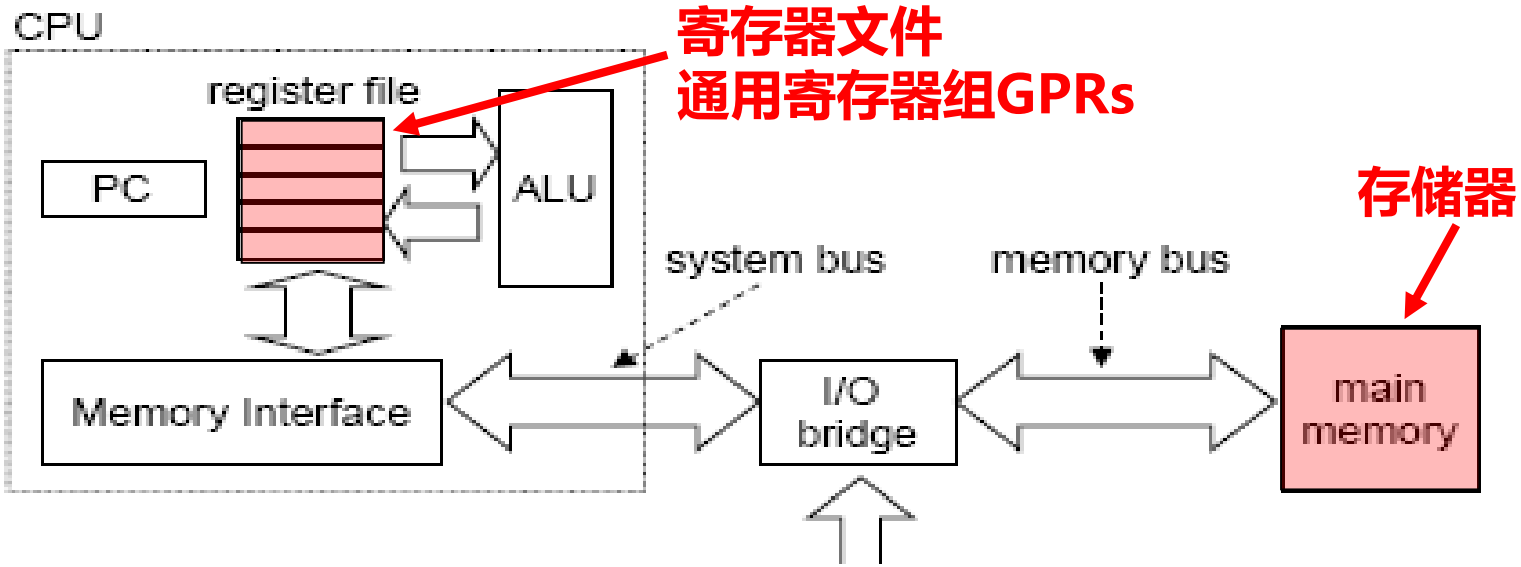
R：寄存器内容

M：存储单元内容

寄存器传送语言 RLT（Register Transfer Language）

计算机中数据的存储

- **计算机中的数据存放在哪里？**



指令中需给出的信息：

操作性质（操作码）

源操作数1 或/和 源操作数2　（立即数、寄存器编号、存储地址）

目的操作数地址　（寄存器编号、存储地址）

存储地址的描述与操作数的数据结构有关！

指令集体系结构ISA

- ISA（Instruction Set Architecture）位于软件和硬件之间
- 硬件的功能通过ISA提供出来
- 软件通过ISA规定的”指令”使用硬件
- ISA规定了：
  - 可执行的指令的集合，包括指令格式、操作种类以及每种操作对应的操作数的相应规定；
  - 指令可以接受的操作数的类型；
  - 操作数所能存放的寄存器组的结构，包括每个寄存器的名称、编号、长度和用途；
  - 操作数所能存放的存储空间的大小和编址方式；
  - 操作数在存储空间存放时按照大端还是小端方式存放；
  - 指令获取操作数的方式，即寻址方式；
  - 指令执行过程的控制方式，包括程序计数器、条件码定义等。

Today: Machine Programming I: Basics

- History of Intel processors and architectures
- **Assembly Basics: Registers, operands, move**
- Arithmetic & logical operations
- C, assembly, machine code

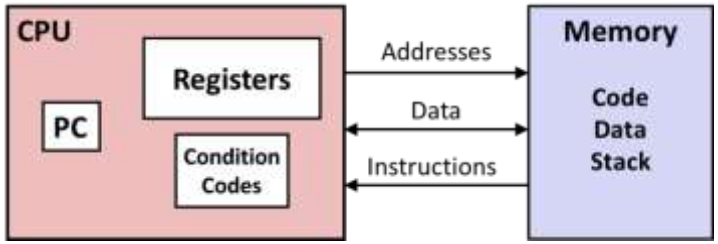
Levels of Abstraction

C programmer

C code

Nice clean layers, but beware…

Assembly programmer



Computer Designer

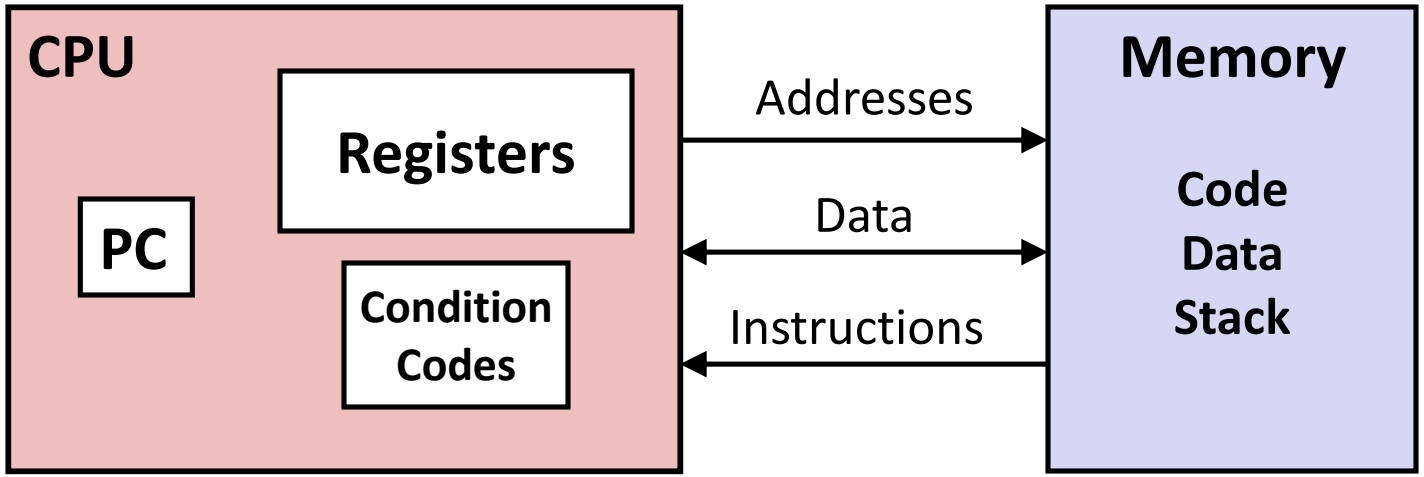
Caches, clock freq, layout, …

Of course, you know that: It's why you are taking this course.

Definitions

- **Architecture: (also ISA: instruction set architecture) The parts of a processor design that one needs to understand for writing correct machine/assembly code**
  - Examples: instruction set specification, registers
  - **Machine Code**: The byte-level programs that a processor executes
  - **Assembly Code**: A text representation of machine code
- **Microarchitecture: Implementation of the architecture**
  - Examples: cache sizes and core frequency
- **Example ISAs:**
  - Intel: x86, IA32, Itanium, x86-64
  - ARM: Used in almost all mobile phones
  - RISC V: New open-source ISA

Assembly/Machine Code View



Programmer-Visible State

- **PC: Program counter**
  - Address of next instruction
  - Called "RIP" (x86-64)
- **Register file**
  - Heavily used program data
- **Condition codes**
  - Store status information about most recent arithmetic or logical operation
  - Used for conditional branching
- **Memory**
  - Byte addressable array
  - Code and user data
  - Stack to support procedures

Assembly Characteristics: Data Types

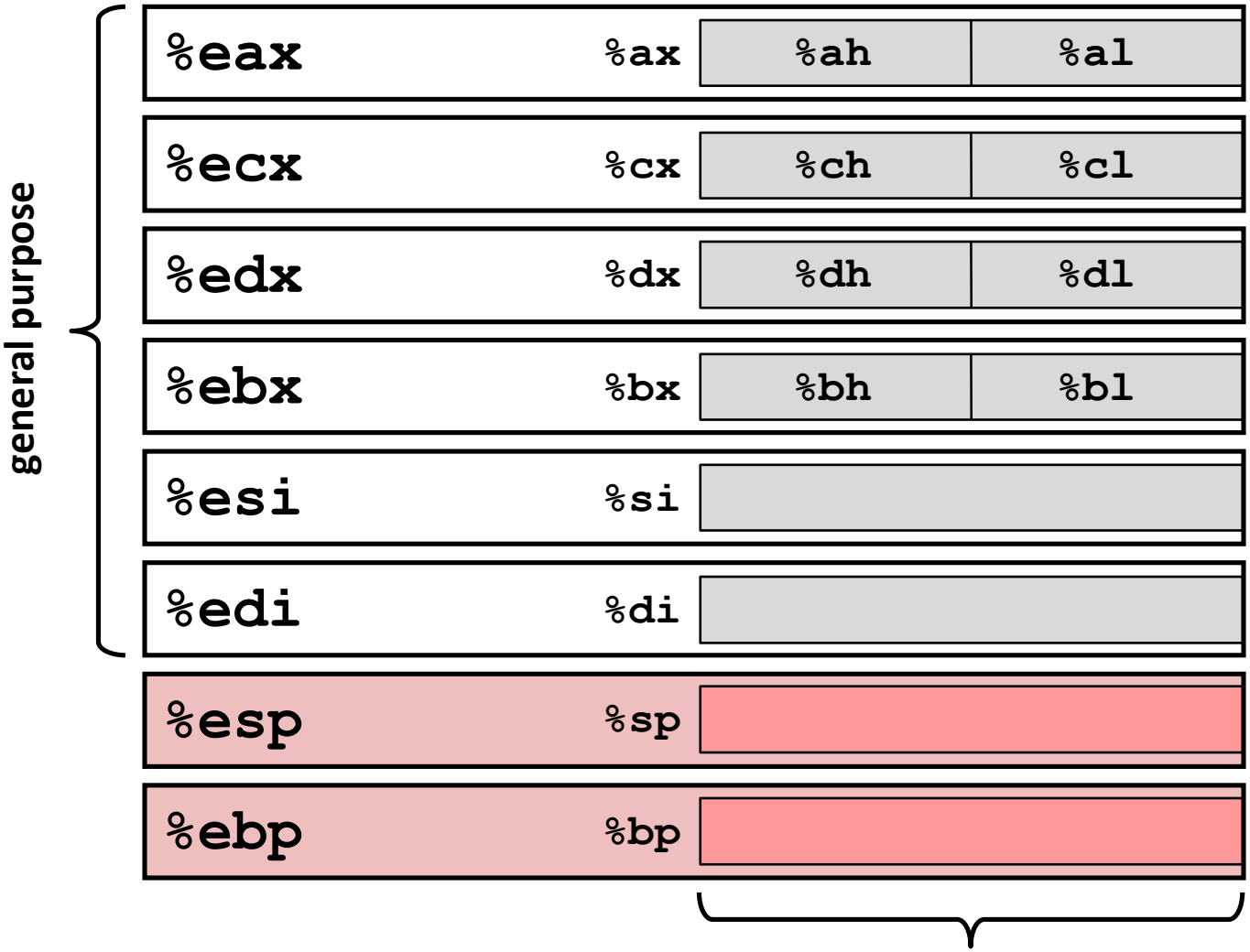
- **"Integer" data of 1, 2, 4, or 8 bytes**
  - Data values
  - Addresses (untyped pointers)
- **Floating point data of 4, 8, or 10 bytes**
- **(SIMD vector data types of 8, 16, 32 or 64 bytes)**
- **Code: Byte sequences encoding series of instructions**
- **No aggregate types such as arrays or structures**
  - Just contiguously allocated bytes in memory

x86-64 Integer Registers

%rax	%eax	%r8	%r8d
%rbx	%ebx	%r9	%r9d
%rcx	%ecx	%r10	%r10d
%rdx	%edx	%r11	%r11d
%rsi	%esi	%r12	%r12d
%rdi	%edi	%r13	%r13d
%rsp	%esp	%r14	%r14d
%rbp	%ebp	%r15	%r15d

- Can reference low-order 4 bytes (also low-order 1 & 2 bytes)
- Not part of memory (or cache)

Some History: IA32 Registers



Assembly Characteristics: Operations

- **Transfer data between memory and register**
  - Load data from memory into register
  - Store register data into memory
- **Perform arithmetic function on register or memory data**
- **Transfer control**
  - Unconditional jumps to/from procedures
  - Conditional branches
  - Indirect branches

IA-32的寻址方式

- 寻址方式
  - 根据指令给定信息得到操作数或操作数地址
- 操作数所在的位置
  - 指令中：立即寻址
  - 寄存器中：寄存器寻址
  - 存储单元中（属于存储器操作数，按字节编址）：其他寻址方式
- 存储器操作数的寻址方式与微处理器的工作模式有关
  - 两种工作模式：实地址模式和保护模式
- 实地址模式（基本用不到）
  - 为与8086/8088兼容而设，加电或复位时
  - 寻址空间为1MB，20位地址：(CS)<<4+(IP)
- 保护模式（需要掌握）
  - 加电后进入，采用虚拟存储管理，多任务情况下隔离、保护
  - 80286以上高档微处理器最常用的工作模式
  - 寻址空间为2^{32}B，32位线性地址分段（段基址+段内偏移量）

保护模式下的寻址方式

寻址方式	说明
立即寻址	指令直接给出操作数
寄存器寻址	指定的寄存器R的内容为操作数
位移	LA=(SR)+A
基址寻址	LA=(SR)+(B)
基址加位移	LA=(SR)+(B)+A
比例变址加位移	LA=(SR)+(I)×S + A
基址加变址加位移	LA=(SR)+(B)+(I)+A
基址加比例变址加位移	LA=(SR)+(B)+(I)×S + A
相对寻址	LA=(PC)+A

存储器操作数

跳转目标指令地址

注：LA:线性地址　(X):X的内容　SR:段寄存器　PC:程序计数器　R:寄存器
A:指令中给定地址段的位移量　B:基址寄存器　I:变址寄存器　S:比例系数

- SR段寄存器（间接）确定操作数所在段的段基址
- 有效地址给出操作数在所在段的偏移地址

存储器操作数的寻址方式(32位)

```
int x;
float a[100];
short b[4][4];
char c;
long *c1;
double d[10];
```

a[i]的地址如何计算？

104+i×4

i=99时，104+99×4=500

b[i][j]的地址如何计算？

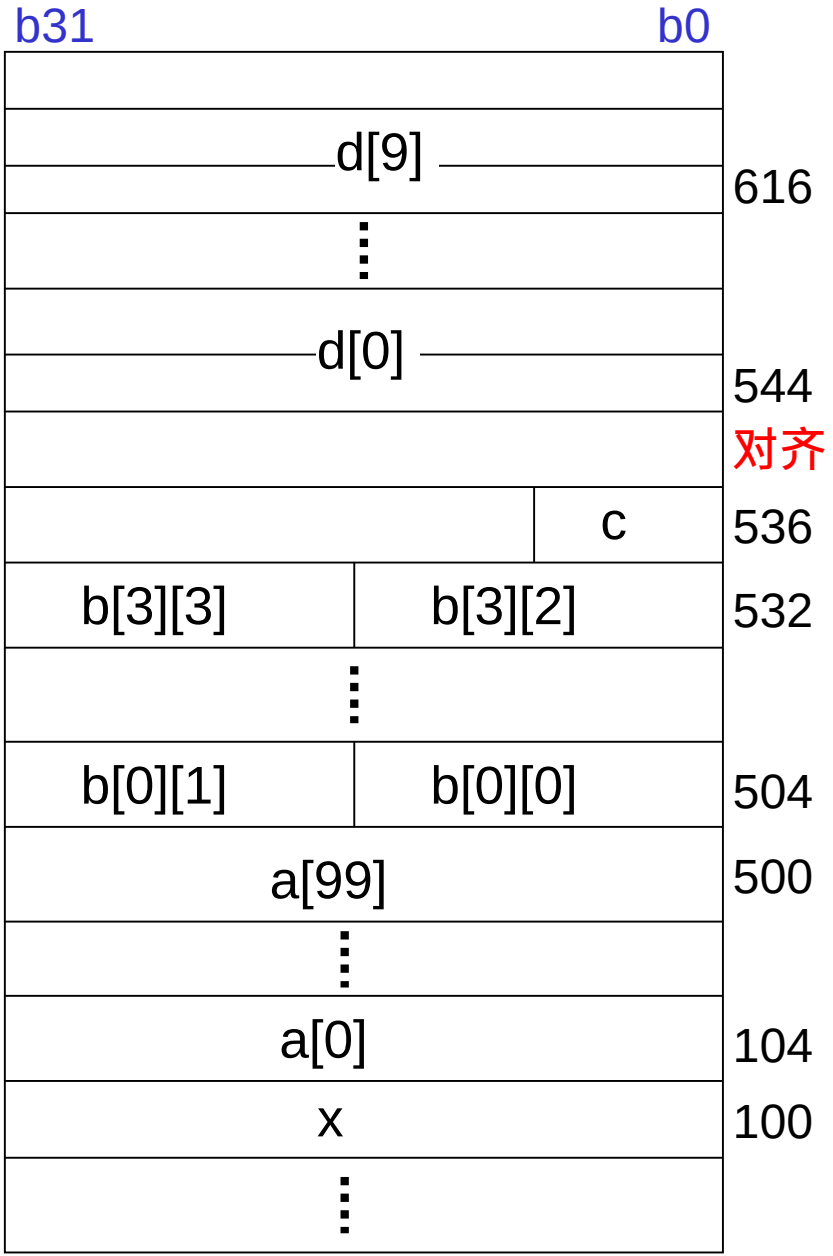
504+i×8+j×2

i=3、j=2时，504+24+4=532

d[i]的地址如何计算？

544+i×8

i=9时，544+9×8=616



存储器操作数的寻址方式

```
int x;
float a[100];
short b[4][4];
char c;
long *c1;
double d[10];
```

各变量应采用什么寻址方式？

x、c：位移 / 基址

a[i]：104+i×4，比例变址+位移

d[i]：544+i×8，比例变址+位移

b[i][j]： 504+i×8+j×2，

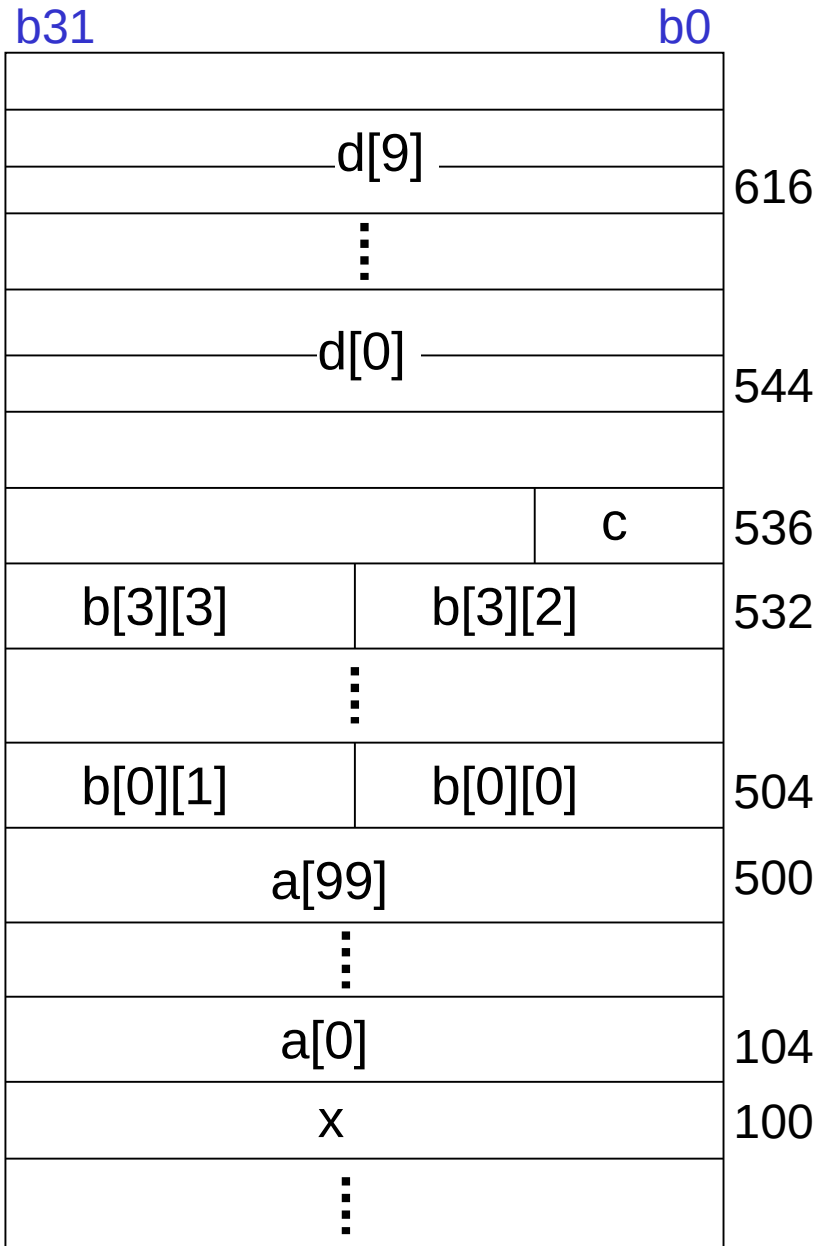
基址+比例变址+位移

将b[i][j]取到EAX中的指令可以是：

“movw 504(%ebp,%esi,2), %eax”

其中， i×8在EBP中，j在ESI中，

2为比例因子



Moving Data

ATT格式

- **Moving Data**

  `movq` *Source, Dest*

C， C++规定， 16进制数必须以 0x开头

- **Operand Types**
  - ***Immediate:*** Constant integer data
    - Example: `$0x400, $-533`
    - Like C constant, but prefixed with `'$'`
    - Encoded with **1, 2, or 4** bytes
  - ***Register:*** One of 16 integer registers
    - Example: `%rax, %r13`
    - But `%rsp` reserved for special use
    - Others have special uses for particular instructions
  - ***Memory:*** 8 consecutive bytes of memory at address given by register
    - Simplest example: `(%rax)`
    - Various other "addressing modes"

`%rax`
`%rcx`
`%rdx`
`%rbx`
`%rsi`
`%rdi`
`%rsp`
`%rbp`

`%rn(n=8~15)`

Warning: Intel docs use mov *Dest, Source*

movq Operand Combinations

	Source	Dest	Src,Dest	C Analog
movq	*Imm*	*Reg*	`movq $0x4,%rax`	`temp = 0x4;`
		Mem	`movq $-147,(%rax)`	`*p = -147;`
	Reg	*Reg*	`movq %rax,%rdx`	`temp2 = temp1;`
		Mem	`movq %rax,(%rdx)`	`*p = temp;`
	Mem	*Reg*	`movq (%rax),%rdx`	`temp = *p;`

Cannot do memory-memory transfer with a single instruction

Simple Memory Addressing Modes

- **Normal (R) Mem[Reg[R]]**
  - Register R specifies memory address
  - Aha! Pointer dereferencing in C

```
movq (%rcx),%rax
```

- **Displacement D(R) Mem[Reg[R]+D]**
  - Register R specifies start of memory region
  - Constant displacement D specifies offset

```
movq 8(%rbp),%rdx
```

Example of Simple Addressing Modes

```
void
whatAmI(<type> a, <type> b)
{
      ????
}
```

%rdi

%rsi

```
whatAmI:
    movq    (%rdi), %rax
    movq    (%rsi), %rdx
    movq    %rdx, (%rdi)
    movq    %rax, (%rsi)
    ret
```

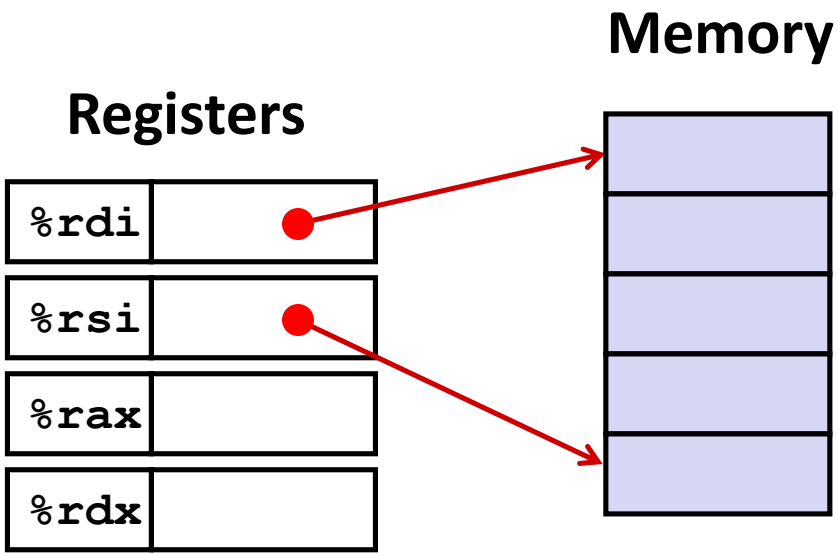
Example of Simple Addressing Modes

```
void swap
   (long *xp, long *yp)
{
  long t0 = *xp;
  long t1 = *yp;
  *xp = t1;
  *yp = t0;
}
```

```
swap:
   movq    (%rdi), %rax
   movq    (%rsi), %rdx
   movq    %rdx, (%rdi)
   movq    %rax, (%rsi)
   ret
```

Understanding Swap()

```
void swap
   (long *xp, long *yp)
{
  long t0 = *xp;
  long t1 = *yp;
  *xp = t1;
  *yp = t0;
}
```



Register	Value
%rdi	xp
%rsi	yp
%rax	t0
%rdx	t1

```
swap:
   movq    (%rdi), %rax   # t0 = *xp
   movq    (%rsi), %rdx   # t1 = *yp
   movq    %rdx, (%rdi)   # *xp = t1
   movq    %rax, (%rsi)   # *yp = t0
   ret
```

Understanding Swap()

Registers

%rdi	0x120
%rsi	0x100
%rax	
%rdx	

Memory

	Address
123	0x120
	0x118
	0x110
	0x108
456	0x100

```
swap:
    movq    (%rdi), %rax   # t0 = *xp
    movq    (%rsi), %rdx   # t1 = *yp
    movq    %rdx, (%rdi)   # *xp = t1
    movq    %rax, (%rsi)   # *yp = t0
    ret
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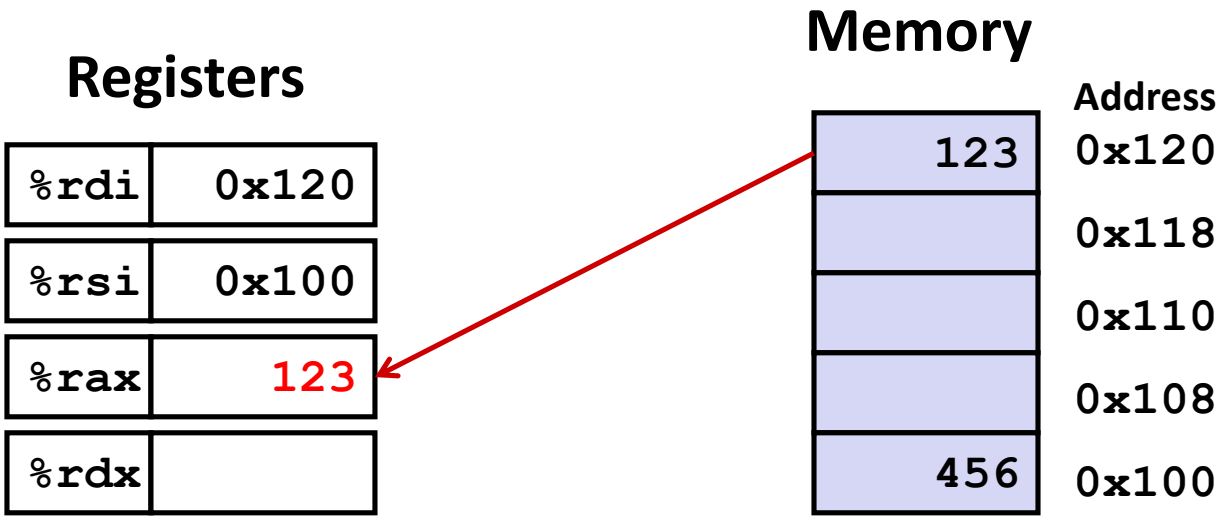
Understanding Swap()

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    movq    %rdx, (%rdi)   # *xp = t1
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    ret
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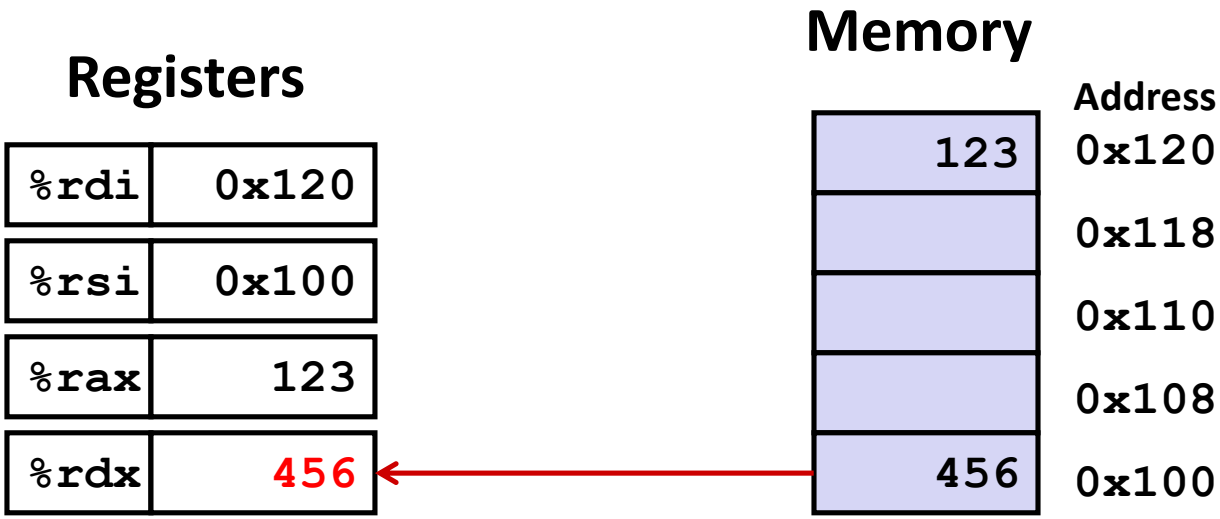
Understanding Swap()

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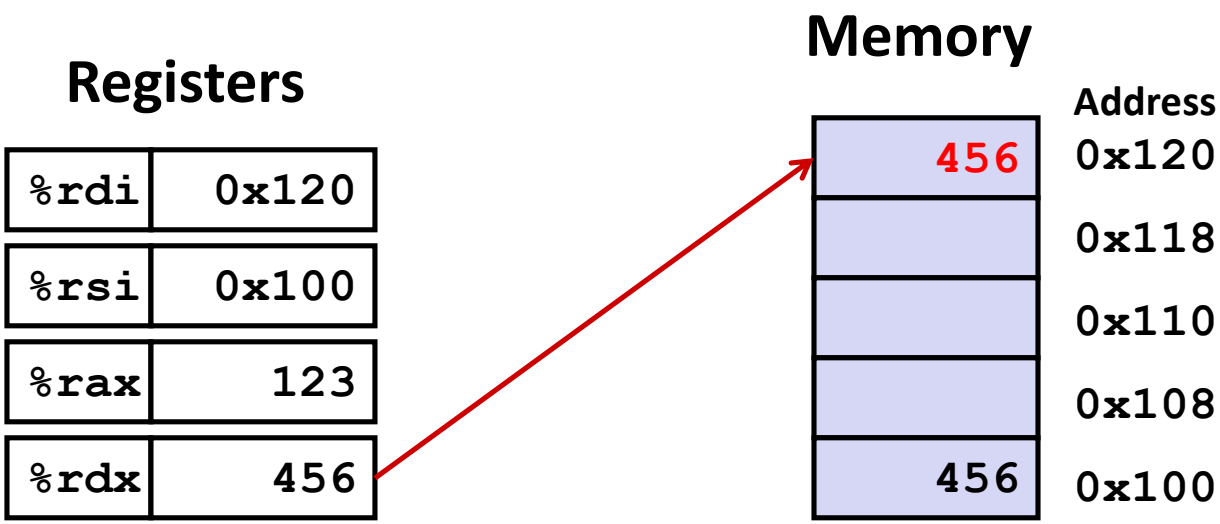
Understanding Swap()

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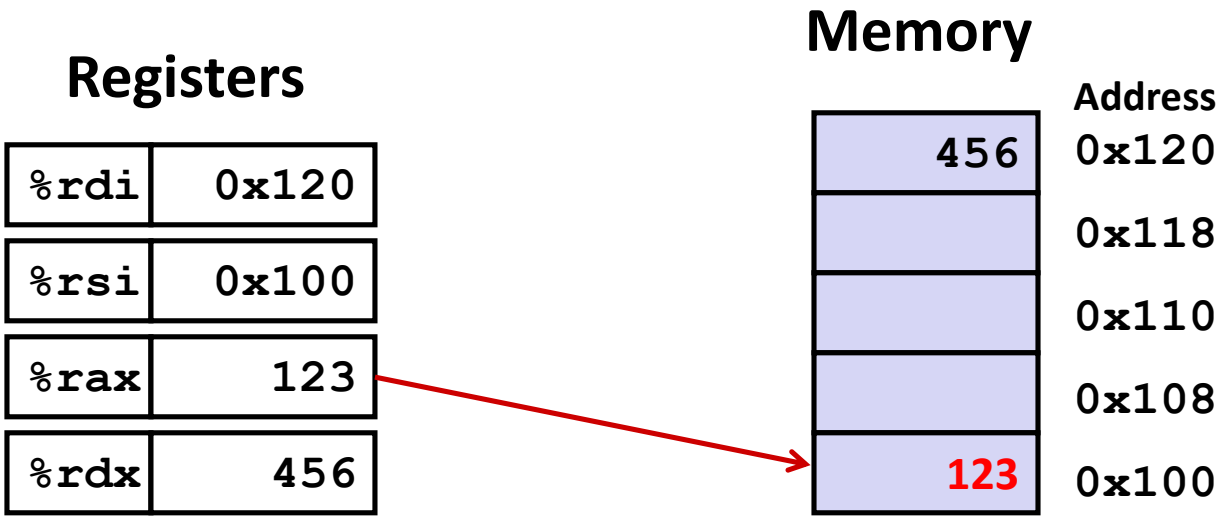
Understanding Swap()

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Simple Memory Addressing Modes

- **Normal (R) Mem[Reg[R]]**
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```
movq (%rcx),%rax
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- **Displacement D(R) Mem[Reg[R]+D]**
  - Register R specifies start of memory region
  - Constant displacement D specifies offset

```
movq 8(%rbp),%rdx
```

Complete Memory Addressing Modes

- **Most General Form**

  D(Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]+ D]**

  - D: Constant "displacement" 1, 2, or 4 bytes
  - Rb: Base register: Any of 16 integer registers
  - Ri: Index register: Any, except for `%rsp`
  - S: Scale: 1, 2, 4, or 8 (*why these numbers?*)

- **Special Cases**

  (Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]]**

  D(Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]+D]**

  (Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]]**

Address Computation Examples

`%rdx`	`0xf000`
`%rcx`	`0x0100`

D(Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]+ D]**

- D: Constant "displacement" 1, 2, or 4 bytes
- Rb: Base register: Any of 16 integer registers
- Ri: Index register: Any, except for `%rsp`
- S: Scale: 1, 2, 4, or 8 (*why these numbers?*)

Expression	Address Computation	Address
`0x8(%rdx)`		
`(%rdx,%rcx)`		
`(%rdx,%rcx,4)`		
`0x80(,%rdx,2)`		

Today: Machine Programming I: Basics

- History of Intel processors and architectures
- Assembly Basics: Registers, operands, move
- **Arithmetic & logical operations**
- C, assembly, machine code

IA-32常用指令类型

（1）传送指令

- 通用数据传送指令

  MOV：一般传送，包括movb、movw和movl等

  MOVS：符号扩展传送，如movsbw、movswl等

  MOVZ：零扩展传送，如movzwl、movzbl等

  XCHG：数据交换

  PUSH/POP：入栈/出栈，如pushl,pushw,popl,popw等

- 地址传送指令

  LEA：加载有效地址，如leal (%edx,%eax), %eax”的功能为 R[eax]←R[edx]+R[eax]，执行前，若R[edx]=*i*，R[eax]=*j*，则指令执行后，R[eax]=*i*+*j*

- 输入输出指令

  IN和OUT：I/O端口与寄存器之间的交换

- 标志传送指令

  PUSHF、POPF：将EFLAG压栈，或将栈顶内容送EFLAG

IA-32常用指令类型

（2）定点算术运算指令

- 加 / 减运算（影响标志、不区分无/带符号）
  - ADD：加，包括addb、addw、addl等
  - SUB：减，包括subb、subw、subl等
- 增1 / 减1运算（影响除CF以外的标志、不区分无/带符号）
  - INC：加，包括incb、incw、incl等
  - DEC：减，包括decb、decw、decl等
- 取负运算（影响标志、若对0取负，则结果为0/CF=0,否则CF=1）
  - NEG：取负，包括negb、negw、negl等
- 比较运算（做减法得到标志、不区分无/带符号）
  - CMP：比较，包括cmpb、cmpw、cmpl等
- 乘 / 除运算（不影响标志、区分无/带符号）
  - MUL / IMUL：无符号乘 / 带符号乘
  - DIV/ IDIV：带无符号除 / 带符号除

IA-32常用指令类型

（3）按位运算指令

- 逻辑运算（仅NOT不影响标志，其他指令OF=CF=0，而ZF和SF根据结果设置：若全0，则ZF=1；若最高位为1，则SF=1 ）
  - NOT：非，包括 notb、notw、notl等
  - AND：与，包括 andb、andw、andl等
  - OR：或，包括 orb、orw、orl等
  - XOR：异或，包括 xorb、xorw、xorl等
  - TEST：做“与”操作测试，仅影响标志
- 移位运算（左/右移时，最高/最低位送CF）
  - SHL/SHR：逻辑左/右移，包括 shlb、shrw、shrl等
  - SAL/SAR：算术左/右移，左移判溢出，右移高位补符
  - ROL/ROR: 循环左/右移，包括 rolb、rorw、roll等
  - RCL/RCR: 带循环左/右移，将CF作为操作数一部分循环移位

以上内容不要死记硬背，遇到具体指令时能查阅到并理解即可。

IA-32常用指令类型

（4）控制转移指令

指令执行可按顺序 或 跳转到转移目标指令处执行

- 无条件转移指令

  JMP DST：无条件转移到目标指令DST处执行

- 条件转移

  Jcc DST：cc为条件码，根据标志（条件码）判断是否满足条件，若满足，则转移到目标指令DST处执行，否则按顺序执行

- 条件设置

  SETcc DST：将条件码cc保存到DST（通常是一个8位寄存器 ）

- 调用和返回指令 （用于过程调用）

  CALL DST：返回地址RA入栈，转DST处执行

  RET：从栈中取出返回地址RA，转到RA处执行

以上内容不要死记硬背，遇到具体指令时能查阅到并理解即可。

Address Computation Instruction

- **`leaq` *Src, Dst***
  - *Src* is address mode expression
  - Set *Dst* to address denoted by expression

- **Uses**
  - Computing addresses without a memory reference
    - E.g., translation of `p = &x[i];`
  - Computing arithmetic expressions of the form x + k*y
    - k = 1, 2, 4, or 8

- **Example**

```
long m12(long x)
{
  return x*12;
}
```

```
leaq ($0x1000), %rax        # %rax = 1000H
leaq (%rbx), %rax           # %rax = %rbx
```

Converted to ASM by compiler:

```
leaq (%rdi,%rdi,2), %rax    # t = x+2*x
salq $2, %rax               # return t<<2
```

Some Arithmetic Operations

- **Two Operand Instructions:**

Format	*Computation*		
`addq`	*Src,Dest*	Dest = Dest + Src	
`subq`	*Src,Dest*	Dest = Dest – Src	
`imulq`	*Src,Dest*	Dest = Dest * Src	
`salq`	*Src,Dest*	Dest = Dest << Src	***Also called shlq***
`sarq`	*Src,Dest*	Dest = Dest >> Src	***Arithmetic***
`shrq`	*Src,Dest*	Dest = Dest >> Src	***Logical***
`xorq`	*Src,Dest*	Dest = Dest ^ Src	
`andq`	*Src,Dest*	Dest = Dest & Src	
`orq`	*Src,Dest*	Dest = Dest | Src	

- **Watch out for argument order! *Src,Dest***
  (Warning: Intel docs use "op *Dest,Src*")
- **No distinction between signed and unsigned int (why?)**

Some Arithmetic Operations

- **One Operand Instructions**

`incq`	*Dest*	*Dest = Dest + 1*
`decq`	*Dest*	*Dest = Dest – 1*
`negq`	*Dest*	*Dest = – Dest*
`notq`	*Dest*	*Dest = ~Dest*

- **See book for more instructions**

Arithmetic Expression Example

```
long arith
(long x, long y, long z)
{
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax
  addq    %rdx, %rax
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx
  leaq    4(%rdi,%rdx), %rcx
  imulq   %rcx, %rax
  ret
```

Interesting Instructions

- **`leaq`**: address computation
- **`salq`**: shift
- **`imulq`**: multiplication
  - But, only used once

Understanding Arithmetic Expression Example

```
long arith
(long x, long y, long z)
{
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax   # t1
  addq    %rdx, %rax          # t2
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx            # t4
  leaq    4(%rdi,%rdx), %rcx  # t5
  imulq   %rcx, %rax          # rval
  ret
```

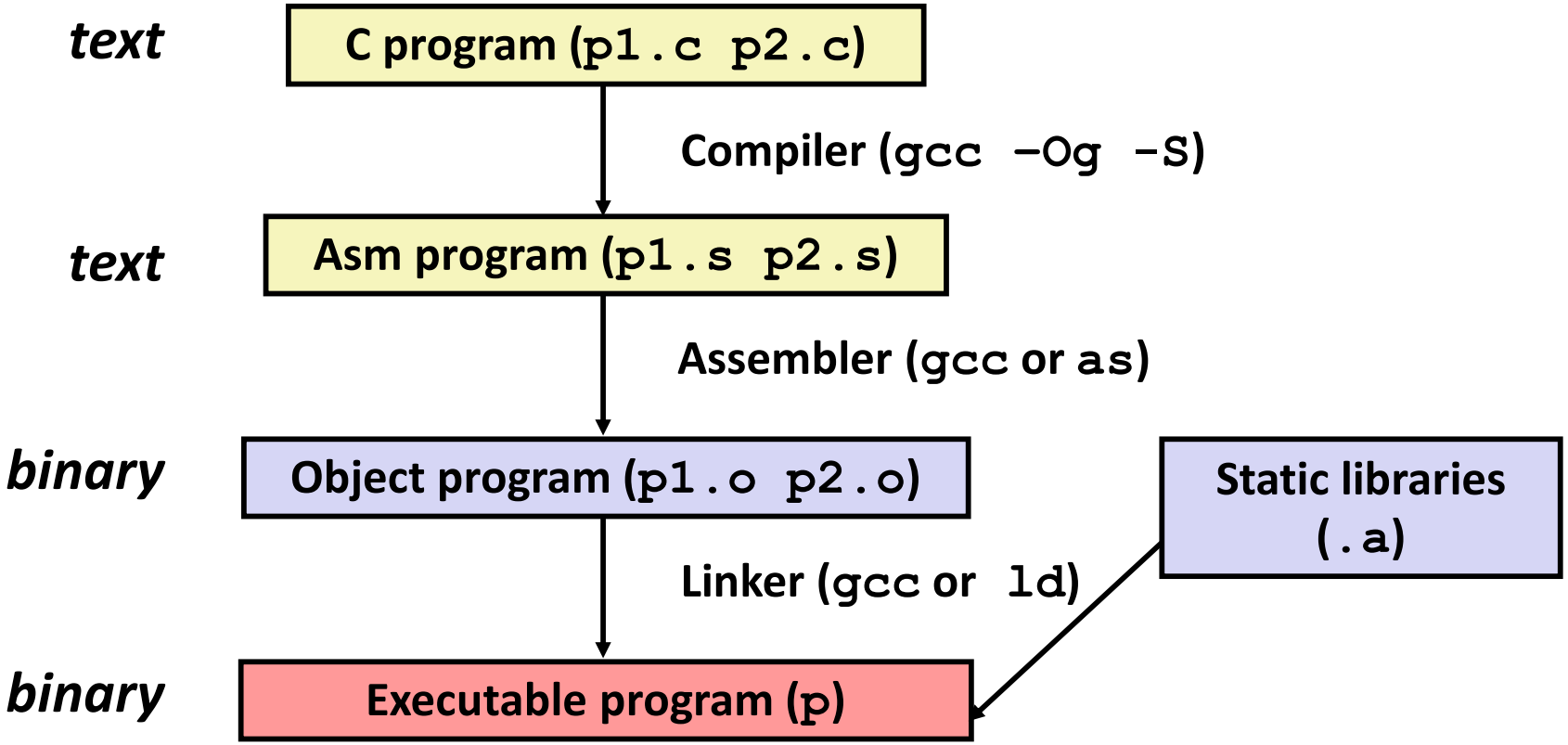
Register	Use(s)
%rdi	Argument x
%rsi	Argument y
%rdx	Argument z, t4
%rax	t1, t2, rval
%rcx	t5

Today: Machine Programming I: Basics

- History of Intel processors and architectures
- Assembly Basics: Registers, operands, move
- Arithmetic & logical operations
- **C, assembly, machine code**

Turning C into Object Code

- Code in files `p1.c p2.c`
- Compile with command: `gcc –Og p1.c p2.c -o p`
  - Use basic optimizations (`–Og`) [New to recent versions of GCC]
  - Put resulting binary in file `p`



Compiling Into Assembly

C Code (sum.c)

```
long plus(long x, long y);

void sumstore(long x, long y,
              long *dest)
{
    long t = plus(x, y);
    *dest = t;
}
```

Generated x86-64 Assembly

```
sumstore:
   pushq   %rbx
   movq    %rdx, %rbx
   call    plus
   movq    %rax, (%rbx)
   popq    %rbx
   ret
```

Obtain (on shark machine) with command

```
gcc –Og –S sum.c
```

Produces file `sum.s`

***Warning*: Will get very different results on non-Shark machines (Andrew Linux, Mac OS-X, …) due to different versions of gcc and different compiler settings.**

What it really looks like

```
        .globl  sumstore
        .type   sumstore, @function
sumstore:
.LFB35:
        .cfi_startproc
        pushq   %rbx
        .cfi_def_cfa_offset 16
        .cfi_offset 3, -16
        movq    %rdx, %rbx
        call    plus
        movq    %rax, (%rbx)
        popq    %rbx
        .cfi_def_cfa_offset 8
        ret
        .cfi_endproc
.LFE35:
        .size   sumstore, .-sumstore
```

What it really looks like

```
        .globl  sumstore
        .type   sumstore, @function
sumstore:
.LFB35:
        .cfi_startproc
        pushq   %rbx
        .cfi_def_cfa_offset 16
        .cfi_offset 3, -16
        movq    %rdx, %rbx
        call    plus
        movq    %rax, (%rbx)
        popq    %rbx
        .cfi_def_cfa_offset 8
        ret
        .cfi_endproc
.LFE35:
        .size   sumstore, .-sumstore
```

Things that look weird and are preceded by a '.' are generally directives.

```
sumstore:
   pushq   %rbx
   movq    %rdx, %rbx
   call    plus
   movq    %rax, (%rbx)
   popq    %rbx
   ret
```

GCC使用举例

```
// test.c
#include <stdio.h>
int add(int i, int j )
{
    int x = i +j;
    return x;
}
```

:test1.c和test2.c，最终生成可执行文件为test

c test2.c -o test

级优化，-O2为二级优化，选项-o指出输出文件名

'objdump -d test.o" 反汇编为汇编语言程序

gcc -E test.c -o test.i

gcc -S test.i -o test.s

gcc –S test.c –o test.s

test.s

```
add:
pushl %ebp
movl  %esp, %ebp
subl  $16, %esp
movl  12(%ebp), %eax
movl  8(%ebp), %edx
leal  (%edx, %eax), %eax
movl  %eax, -4(%ebp)
movl  -4(%ebp), %eax
leave
ret
```

00000000 <add>:

位移量	机器指令	汇编指令
0:	55	push %ebp
1:	89 e5	mov %esp, %ebp
3:	83 ec 10	sub $0x10, %esp
6:	8b 45 0c	mov 0xc(%ebp), %eax
9:	8b 55 08	mov 0x8(%ebp), %edx
c:	8d 04 02	lea (%edx,%eax,1), %eax
f:	89 45 fc	mov %eax, -0x4(%ebp)
12:	8b 45 fc	mov -0x4(%ebp), %eax
15:	c9	leave
16:	c3	ret

编译得到的与反汇编得到的汇编指令形式稍有差异

两种目标文件

```c
// test.c
#include <stdio.h>
int add(int i, int j )
{
    int x = i +j;
    return x;
}
```

test.o：可重定位目标文件

test：可执行目标文件

“objdump -d test.o” 结果

00000000 <add>:

```
0:   55          push   %ebp
1:   89 e5       mov    %esp, %ebp
3:   83 ec 10    sub    $0x10, %esp
6:   8b 45 0c    mov    0xc(%ebp), %eax
9:   8b 55 08    mov    0x8(%ebp), %edx
c:   8d 04 02    lea    (%edx,%eax,1), %eax
f:   89 45 fc    mov    %eax, -0x4(%ebp)
12:  8b 45 fc    mov    -0x4(%ebp), %eax
15:  c9          leave
16:  c3          ret
```

“objdump -d test” 结果

080483d4 <add>:

```
80483d4:  55          push ...
80483d5:  89 e5       …
80483d7:  83 ec 10    …
80483da:  8b 45 0c    …
80483dd:  8b 55 08    …
80483e0:  8d 04 02    …
80483e3:  89 45 fc    …
80483e6:  8b 45 fc    …
80483e9:  c9          …
80483ea:  c3          ret
```

test.o中的代码从地址0开始，test中的代码从80483d4开始！

Object Code

Code for `sumstore`

```
0x0400595:
    0x53
    0x48
    0x89
    0xd3
    0xe8
    0xf2
    0xff
    0xff
    0xff
    0x48
    0x89
    0x03
    0x5b
    0xc3
```

- **Total of 14 bytes**
- **Each instruction 1, 3, or 5 bytes**
- **Starts at address `0x0400595`**

- **Assembler**
  - Translates `.s` into `.o`
  - Binary encoding of each instruction
  - Nearly-complete image of executable code
  - Missing linkages between code in different files
- **Linker**
  - Resolves references between files
  - Combines with static run-time libraries
    - E.g., code for **`malloc`**, **`printf`**
  - Some libraries are *dynamically linked*
    - Linking occurs when program begins execution

Machine Instruction Example

```
*dest = t;
```

```
movq %rax, (%rbx)
```

```
0x40059e:  48 89 03
```

- **C Code**
  - Store value **`t`** where designated by **`dest`**
- **Assembly**
  - Move 8-byte value to memory
    - Quad words in x86-64 parlance
  - Operands:

    `t`: Register **`%rax`**

    `dest`: Register **`%rbx`**

    `*dest`: Memory **M[`%rbx`]**
- **Object Code**
  - 3-byte instruction
  - Stored at address **`0x40059e`**

Disassembling Object Code

Disassembled

```
0000000000400595 <sumstore>:
  400595:  53                  push   %rbx
  400596:  48 89 d3            mov    %rdx,%rbx
  400599:  e8 f2 ff ff ff      callq  400590 <plus>
  40059e:  48 89 03            mov    %rax,(%rbx)
  4005a1:  5b                  pop    %rbx
  4005a2:  c3                  retq
```

- **Disassembler**

  `objdump -d sum`

  - Useful tool for examining object code
  - Analyzes bit pattern of series of instructions
  - Produces approximate rendition of assembly code
  - Can be run on either `a.out` (complete executable) or `.o` file

Alternate Disassembly

Disassembled

```
Dump of assembler code for function sumstore:
 0x0000000000400595 <+0>: push   %rbx
 0x0000000000400596 <+1>: mov    %rdx,%rbx
 0x0000000000400599 <+4>: callq  0x400590 <plus>
 0x000000000040059e <+9>: mov    %rax,(%rbx)
 0x00000000004005a1 <+12>:pop    %rbx
 0x00000000004005a2 <+13>:retq
```

- **Within gdb Debugger**
  - Disassemble procedure

```
gdb sum
disassemble sumstore
```

Alternate Disassembly

Object Code

```
0x0400595:
    0x53
    0x48
    0x89
    0xd3
    0xe8
    0xf2
    0xff
    0xff
    0xff
    0x48
    0x89
    0x03
    0x5b
    0xc3
```

Disassembled

```
Dump of assembler code for function sumstore:
 0x0000000000400595 <+0>: push   %rbx
 0x0000000000400596 <+1>: mov    %rdx,%rbx
 0x0000000000400599 <+4>: callq  0x400590 <plus>
 0x000000000040059e <+9>: mov    %rax,(%rbx)
 0x00000000004005a1 <+12>:pop    %rbx
 0x00000000004005a2 <+13>:retq
```

- **Within gdb Debugger**
  - Disassemble procedure

    `gdb sum`

    `disassemble sumstore`
  - Examine the 14 bytes starting at `sumstore`

    `x/14xb sumstore`

What Can be Disassembled?

```
% objdump -d WINWORD.EXE

WINWORD.EXE:   file format pei-i386

No symbols in "WINWORD.EXE".
Disassembly of section .text:

30001000 <.text>:
30001000:
30001001:
30001003:
30001005:
3000100a:
```

Reverse engineering forbidden by Microsoft End User License Agreement

- **Anything that can be interpreted as executable code**
- **Disassembler examines bytes and reconstructs assembly source**

Machine Programming I: Summary

- **History of Intel processors and architectures**
  - Evolutionary design leads to many quirks and artifacts
- **C, assembly, machine code**
  - New forms of visible state: program counter, registers, ...
  - Compiler must transform statements, expressions, procedures into low-level instruction sequences
- **Assembly Basics: Registers, operands, move**
  - The x86-64 move instructions cover wide range of data movement forms
- **Arithmetic**
  - C compiler will figure out different instruction combinations to carry out computation